Apache Spark installation:

Spark Standalone Mode → https://spark.apache.org/docs/latest/spark-standalone.html

Spark 3.2.0 Overview Programming Guides

Spark Standalone Mode

- Security
- Installing Spark Standalone to a Cluster
- Starting a Cluster Manually
- Cluster Launch Scripts
- Resource Allocation and Configuration Overview
- Connecting an Application to the Cluster
- Client Properties
- Launching Spark Applications
- Resource Scheduling
- Executors Scheduling

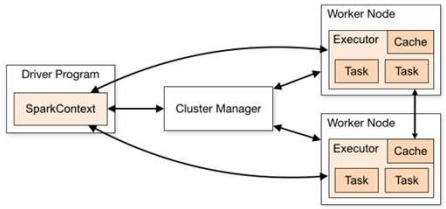
Cluster Mode Overview

This document gives a short overview of how Spark runs on clusters, to make it easier to understand the components involved. Read through the application submission guide to learn about launching applications on a cluster.

Components

Spark applications run as independent sets of processes on a cluster, coordinated by the SparkContext object in your main program (called the *driver program*).

Specifically, to run on a cluster, the SparkContext can connect to several types of *cluster managers* (either Spark's own standalone cluster manager, Mesos, YARN or Kubernetes), which allocate resources across applications. Once connected, Spark acquires *executors* on nodes in the cluster, which are processes that run computations and store data for your application. Next, it sends your application code (defined by JAR or Python files passed to SparkContext) to the executors. Finally, SparkContext sends *tasks* to the executors to run.



Cluster Mode Overview → https://spark.apache.org/docs/latest/cluster-overview.html

Running Spark on YARN → https://spark.apache.org/docs/latest/running-on-yarn.html#configuration

Spark 3.2.0 Overview Programming Guides

Running Spark on YARN

- Security
- Launching Spark on YARN
  - Adding Other JARs
- Preparations
- Configuration
- Debugging your Application
  - Spark Properties
  - Available patterns for SHS custom executor log URL
- Resource Allocation and Configuration Overview
- Stage Level Scheduling Overview
- Important notes
- Kerberos
  - YARN-specific Kerberos Configuration
  - Troubleshooting Kerberos

Standalone Apache Spark Installation:

- JDK installation – version v21
- Installation of Spark itself and all its many dependencies and env variables.
- Scala IDE installation

Standalone Apache Spark Installation:

A.

After downloading and installing the JDK → set the environment variables:

Then set the environment variables:

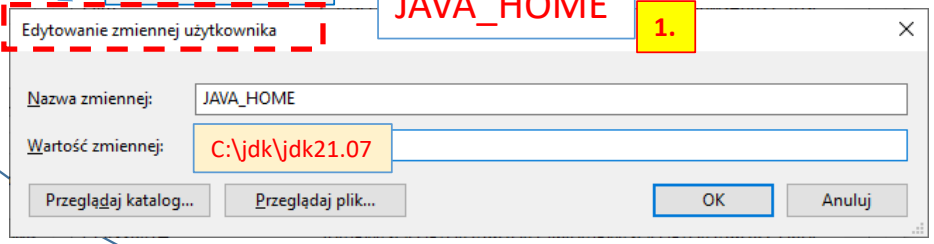
it should be unpacked to a path without spaces:

C:\jdk\jdk21.07

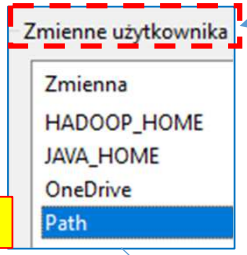
user variable

JAVA_HOME

1.

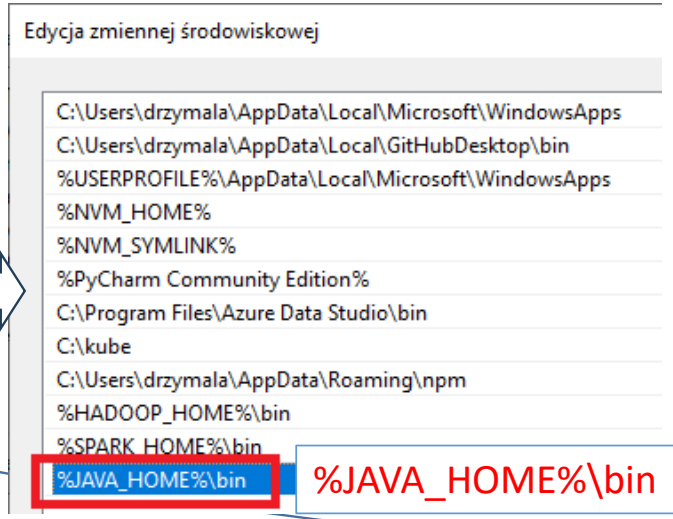


user variable



2.

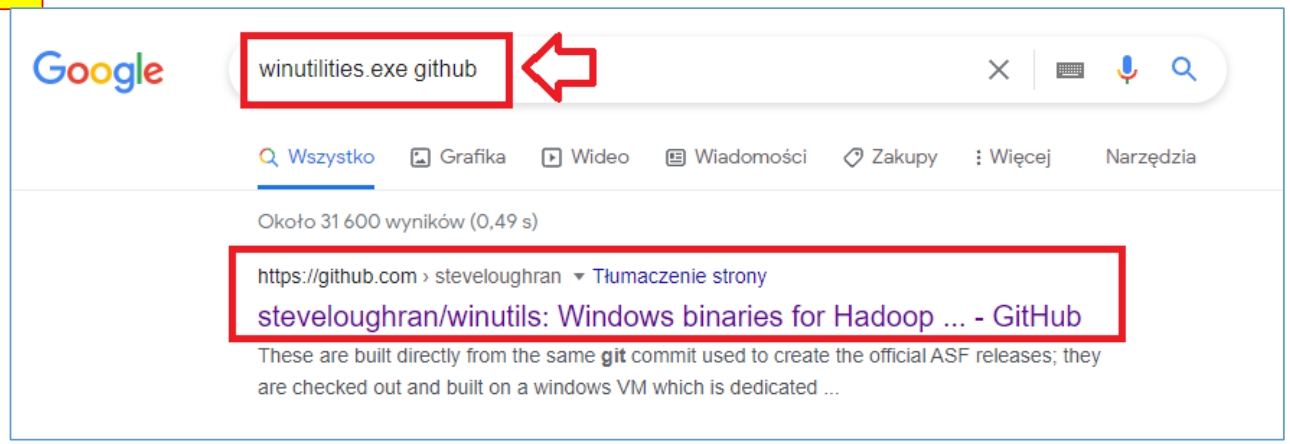
PATH



%JAVA_HOME%\bin

B.

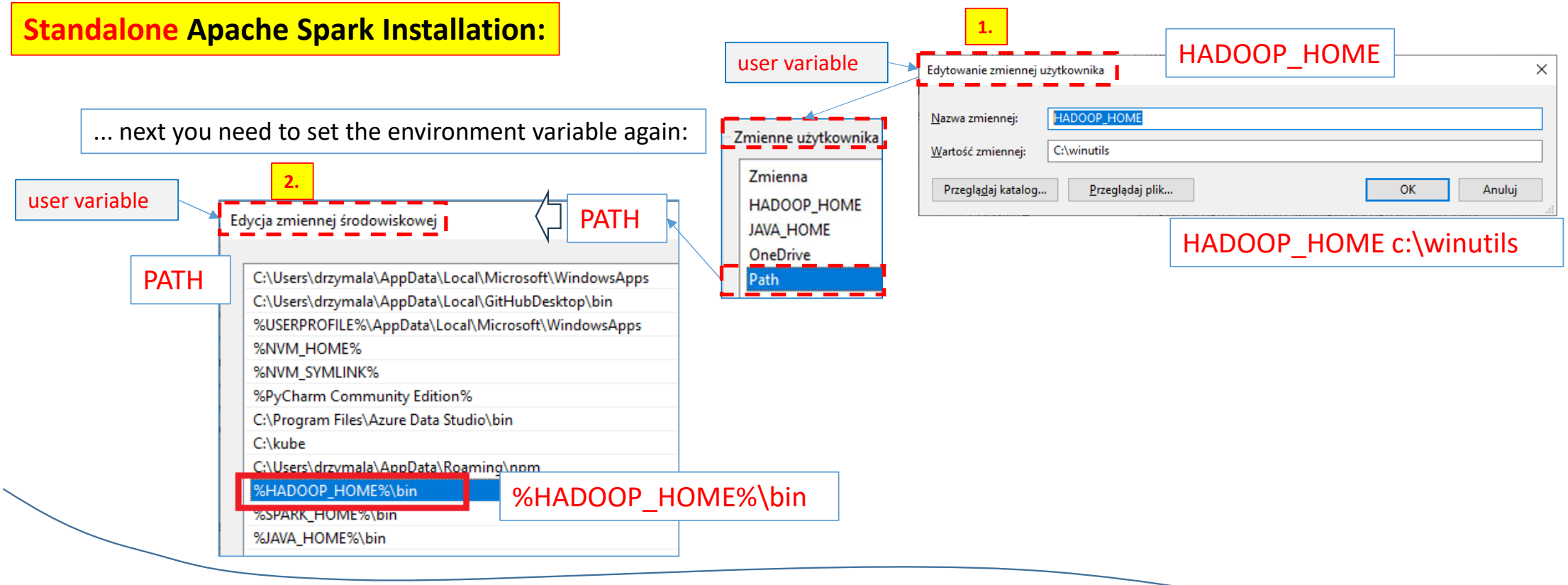
Download: winutilities.exe from WIKAMP (or github for the appropriate version)



... and copying it to a folder on drive c:\

C:\winutils\bin

Standalone Apache Spark Installation:



C.

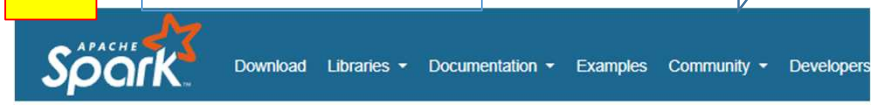
Create a directory: → **c:\BigData\hive**

Go to the directory with the command: → **cd c:\winutils\bin**

Run the command: → **winutils.exe chmod 777 c:\BigData\hive**

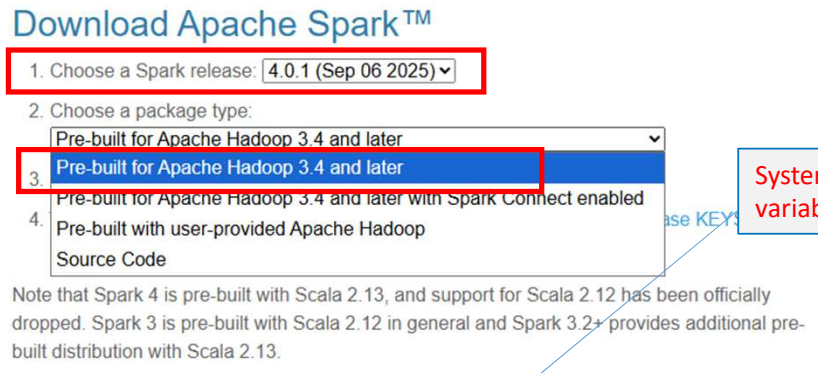
Standalone Apache Spark Installation:

D. ... and install Spark:



https://spark.apache.org/downloads.html

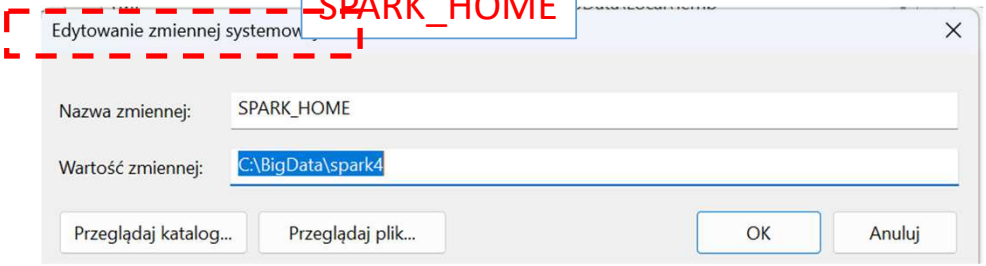
... and after unpacking, copy it to the directory: C:\BigData\spark4



... we set the environment variable again:

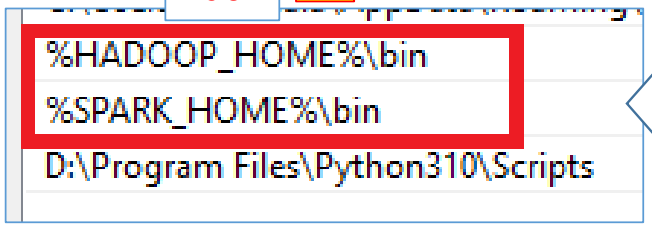
System (or user) variable **1.**

SPARK_HOME



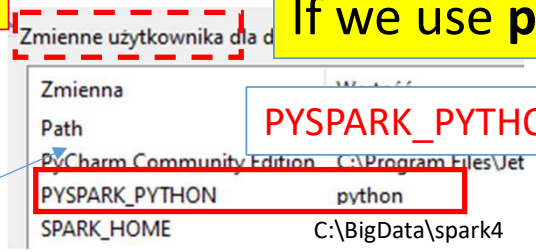
... and path:

Path **2.**



%SPARK_HOME%\bin

user variable **3.**



If we use **pySpark**: !!!!

PYSPARK_PYTHON → python

ATTENTION!!! Just

Only a given user can run pySpark

Standalone Apache Spark Installation:

E.

Open folder: **c:\BigData\spark4\conf** (make sure "Filename extensions" is checked)

Rename the file: **log4j.properties.template** for: **log4j.properties**



Edit this file (using notepad) and change the **error** level from **INFO** to **ERROR** for **log4j**.rootCategory

... and you can write scripts in any IDE: PyCharm or e.g. Eclipse

How you can Install Apache Spark with Python (PySpark)?:

…you should issue the command:

python -m pip install pyspark

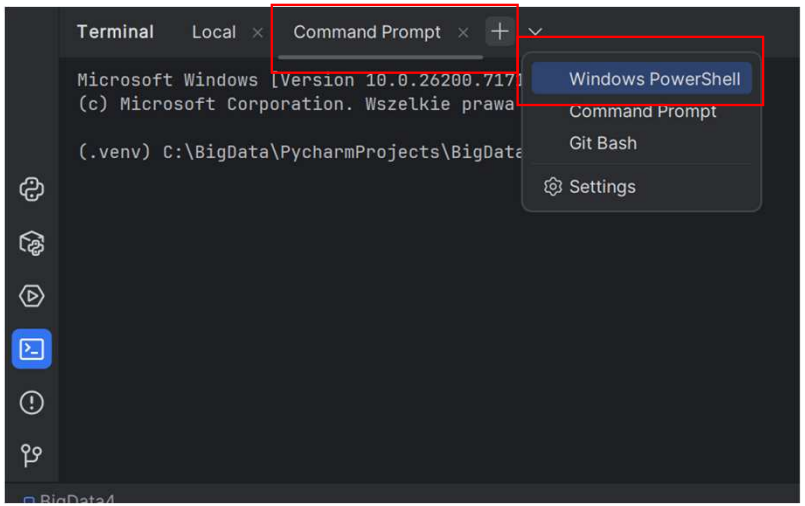
The ~ca: 281MB package is installed

```
D:\Program Files\Python310>python -m pip install pyspark
Collecting pyspark
  Downloading pyspark-3.2.0.tar.gz (281.3 MB)
     |████████████████████████████████| 281.3 MB 1.9 kB/s
Collecting py4j==0.10.9.2
  Downloading py4j-0.10.9.2-py2.py3-none-any.whl (198 kB)
     |████████████████████████████████| 198 kB 2.2 MB/s
Using legacy 'setup.py install' for pyspark, since package 'wheel' is not installed.
Installing collected packages: py4j, pyspark
    Running setup.py install for pyspark ... done
Successfully installed py4j-0.10.9.2 pyspark-3.2.0
WARNING: You are using pip version 21.2.3; however, version 21.3 is available.
You should consider upgrading via the 'D:\Program Files\Python310\python.exe -m pip install --upgrade pip' command.

D:\Program Files\Python310>
```

…and open:

CMD



A prompt appears:

…and run:

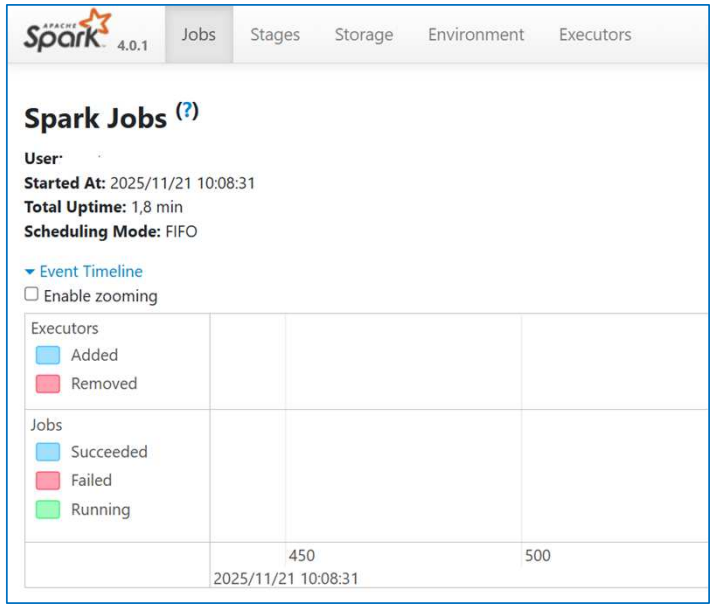
pyspark

```
Type "help", "copyright", "credits" or "license" for more information.
WARNING: Using incubator modules: jdk.incubator.vector
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
Welcome to
      ____              __
     / __/__  ___ _____/ /__
    _\ \/ _ \/ _ `/ __/  '_/
   /__ / .__/\_,_/_/ /_/\_\   version 4.0.1
      /_/

n 3.13.7 (tags/v3.13.7:bcee1c3, Aug 14 2025 14:15:11)
I  le at http://host.docker.internal:4040
Spark context available a  'sc' (master = local[*], app id = local-1763716111912).
SparkSession available as 'spark'.
>>>
```

…and open:

How you can Install Apache Spark with Python (PySpark)?:



http://host.docker.internal:4040/jobs/

Executors

Show Additional Metrics

Summary

	RDD Blocks	Storage Memory	Disk Used	Cores	Active Tasks	Failed Tasks	Complete Tasks	Total Tasks	Task Time (GC Time)	Input	Shuffle Read	Shuffle Write
Active(1)	0	0.0 B / 434.4 MiB	0.0 B	32	0	0	0	0	7.1 min (35.0 ms)	0.0 B	0.0 B	0.0 B
Dead(0)	0	0.0 B / 0.0 B	0.0 B	0	0	0	0	0	0.0 ms (0.0 ms)	0.0 B	0.0 B	0.0 B
Total(1)	0	0.0 B / 434.4 MiB	0.0 B	32	0	0	0	0	7.1 min (35.0 ms)	0.0 B	0.0 B	0.0 B

Executors

Show 20 entries

Executor ID	Address	Status	RDD Blocks	Storage Memory	Disk Used	Cores	Active Tasks	Failed Tasks	Complete Tasks	Total Tasks	Task Time (GC Time)	Input	Shuffle Read	Shuffle Write	Thread Dump	Heap Histogram	Add Time
driver	host.docker.internal:54392	Active	0	0.0 B / 434.4 MiB	0.0 B	32	0	0	0	0	7.1 min (35.0 ms)	0.0 B	0.0 B	0.0 B	Thread Dump	Heap Histogram	2025-11-21 1

A prompt appears:

ATTENTION !!!

To work with pySpark

install the development environment:

PyCharm Community Edition

https://www.jetbrains.com/pycharm/download/?section=windows



… then create a new project and install the pySpark libraries

ATTENTION !!!

Python 3.12 → Changing package dependencies

(in further methods of the SparkML library → ml an error appears:

Described in the package documentation https://peps.python.org/pep-0632/

ModuleNotFoundError: No module named 'distutils'

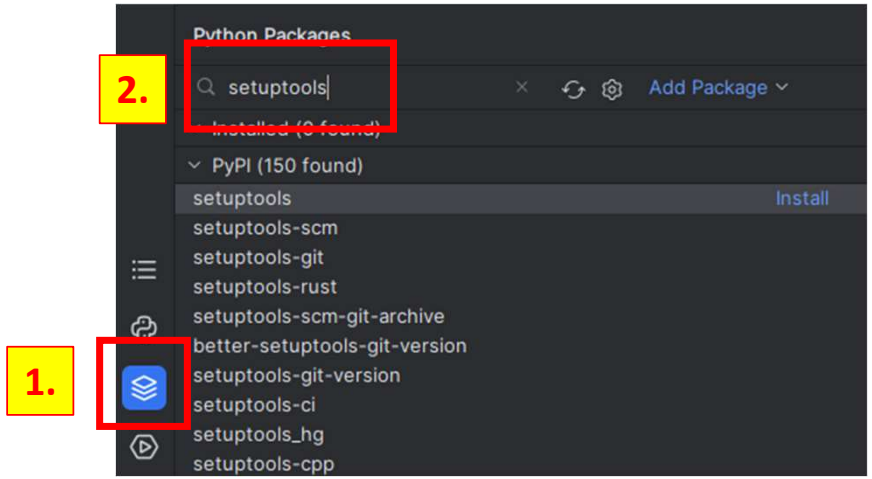
PEP 632 - Deprecated distutils module

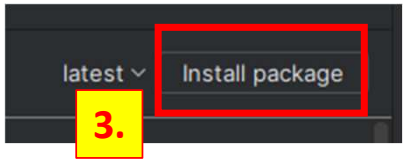
From version 3.12 → install the package:

You don't need to reinstall **pyspark** and **python** → you only need to install **setuptools**

pip install setuptools

or… → environment: pyCharm:





Starting the task:

python.exe .\ml_2.py